

Measurement of electron–hole pair creation energy in wide bandgap semiconductors

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ABSTRACT

The method for the measurements of average energy necessary for the creation of an electron-hole pair in semiconductors based on fitting the collected current dependence on beam energy has been described. The effect of metal film on the backscatter energy is estimated for 4H–SiC, GaN and β -Ga₂O₃ Schottky barriers. The empirical expressions describing the dependence of this energy on beam energy for bulk material and thin film are proposed. The approximated depth-dose dependence for 4H–SiC is obtained by fitting the dependence calculated using the Monte Carlo algorithm. Values of average electron-hole pair ionization energy are estimated for 4H–SiC, GaN and β -Ga₂O₃ taking into account the beam energy absorption in the metal film and its effect on the electron backscattering. The values obtained are 8.04 ± 0.13 , 10 ± 0.5 eV and 14.8 eV for 4H–SiC, GaN and β -Ga₂O₃, respectively.

1. Introduction

The average energy consumed in the creation of an electron-hole pair in semiconductors E_i is an important parameter determining the sensitivity of charged particles and x-ray detectors (De Napoli, 2022; Prasad et al., 2023; Capan, 2024). It also determines the conversion efficiency of semiconductor radioisotope microbatteries (Spencer and Alam, 2019; Yakimov, 2023) and electron-beam-pumped lasers (Gamov et al., 2015; Hayashi et al., 2017). Usually it can be accounted for by a sum of three contributions: semiconductor bandgap, phonon losses (mainly optical phonons) and the residual kinetic energy (Klein, 1968; Kobayashi, 1972; Alig and Bloom, 1975; Lioliou and Barnett, 2022; Yakimov et al., 2021a). The last two terms were estimated by some theoretical considerations and by fitting available experimental data. As a result, the following expressions for E_i as a function of bandgap E_g were proposed: $E_i = 2.8 \times E_g + 0.5$ – 1 eV (Klein, 1968), $E_i = 2.59 \times E_g + 0.71$ eV (Kobayashi, 1972), $E_i = 2.7 \times E_g + 0.55$ eV (Alig and Bloom, 1975). As shown in (Yakimov et al., 2021a), all three relations describe the more recent data rather well, although, as shown in (Lioliou and Barnett, 2022; Whitaker et al., 2020), for many of ternary semiconductors the expression $E_i = 1.54 \times E_g + 1.89$ eV better describes the bandgap dependence of E_i . A difference between the last expression and those proposed in (Klein, 1968; Kobayashi, 1972; Alig and Bloom, 1975) is especially significant for larger bandgap semiconductors. Thus, at present the exact value of E_i can only be obtained experimentally.

Wide- and ultra-wide bandgap semiconductors, such as SiC, GaN and Ga₂O₃, are now considered as promising materials for radiation detectors (De Napoli, 2022; Prasad et al., 2023; Capan, 2024), power devices with the high irradiation resistance and semiconductor converter in radioisotope batteries (Spencer and Alam, 2019). To predict properties of such devices and optimize their design the knowledge of E_i value is required. However, the values of E_i given in the literature for GaN and 4H–SiC differ significantly. Thus, the reported E_i values for 4H–SiC vary from 5.05 (Chandrashekar et al., 2006) to 8.6 eV (Lebedev et al., 2004). The most recent value given in (Gspöner et al., 2024) is 7.83 ± 0.02 eV. The E_i measurements in GaN and Ga₂O₃ are not so numerous. For GaN the reported values are 8.9 (Sellin and Vaitkus, 2006) and 9.59 eV (Yakimov et al., 2021a). For β -Ga₂O₃ E_i was estimated in (Nielsen and Fischetti, 2023) by solving the semiclassical Boltzmann transport equation as 11.24 eV and in (Yakimov et al., 2021a) from the EBIC measurements as 15.6 eV.

As a rule, E_i value is obtained by measuring the collected current induced in a Schottky diode by charged particles or soft x rays whose intensity is carefully measured with a well-calibrated detector with known E_i (Whitaker et al., 2020; Gspöner et al., 2024). For such measurements, it is important to keep the collection efficiency close to unity to avoid incomplete charge collection that leads to E_i overestimation. This is usually achieved by applying a high reverse bias to a detector to enhance carrier transport to the electrodes. When the collection efficiency approaches 100 %, the collected current should be independent

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of applied bias. However, in GaN and Ga₂O₃ the diffusion length is lower than 600 nm (Yakimov, 2015; Yakimov et al., 2018; Polyakov et al., 2024) and in such semiconductors the collection probability can be lower than 100 %, even at high enough bias voltages. Additionally, in many structures based on GaN and Ga₂O₃, the current gain has been observed, even at bias voltages lower than that necessary for avalanche multiplication (Katz et al., 2001; Yakimov et al., 2021b; Polyakov et al., 2025). Any induced current gain mechanism leads to an underestimation of E_i value. Perhaps these two factors are the reason for the large spread of reported E_i values.

As shown in (Yakimov et al., 2021a; Svintsov et al., 2018), the average electron-hole pair ionization energy can be obtained from fitting the collected current dependence in the Electron Beam Current Induced (EBIC) mode on beam energy. As shown in (Yakimov et al., 2021a), the E_i value obtained by this method for Si is in good agreement with the available literature data. The main advantage of the method is the self-consistent determination of the collection probability and its effect on the collected current. Therefore, there is no need to use high voltages to reduce the loss of excess carriers. Moreover, there is no need for a well-calibrated detector to measure the absorbed energy. The main problems leading to errors in the determination of E_i by this method are related to carrier recombination at the metal/semiconductor interface, and in the case of Ga₂O₃ to induced current gain in some Schottky barriers even at zero applied bias. However, the similar problems arise in the traditional method.

The question may arise whether the value of average electron-hole pair ionization energy obtained for low-energy electrons can be used for other types of ionizing radiation (protons, α -particles, X-ray, heavy ions etc) with a wide range of energies. This point has not been considered in depth theoretically. However, many studies have noted that the values experimentally obtained for Si, Ge and SiC using electrons, X-ray, protons and α -particles are similar (Gspöner et al., 2024; Klein, 1967; Scholze et al., 2000). A dependence on the energy of ionizing particles can be also not essential because, as shown in (Fang et al., 2018), that although high-energy particles lose energy by various processes following the initial scattering events, the average energy required to create a thermalized electron-hole pair is determined only by interactions that occur after the carriers have energies of ~ 10 eV or less. As for heavy ions, a significant part of their energy is spent on ionization losses, and it is this mechanism that is predominant for low-energy electrons.

In the present paper the method has been described and analyzed in more details. The effect of metal film on the energy losses by absorption and backscattering is estimated for 4H-SiC, GaN and β -Ga₂O₃ Schottky barriers. The empirical expression describing the dependence of backscatter energy on beam energy is proposed. The values of average electron-hole pair ionization energy are estimated for 4H-SiC, GaN and β -Ga₂O₃ taking into account the beam energy absorption in the metal film and its effect on the electron backscattering.

2. Experimental

The EBIC measurements were carried out in a JSM 840A (Jeol) scanning electron microscope at room temperature with a beam energy E_b varied in the range from 3 to 40 keV and a beam current I_b of 0.1 nA. The current amplifier used was a Keithley 428 amplifier. Beam current is measured with the Faraday cup. For E_i measurements in 4H-SiC the Schottky barriers were prepared on five samples with different dopant concentration and diffusion length values by evaporation of Ni or Au. Additionally, p-n junction with the junction depth of 0.4 μ m was studied. The Schottky barriers on five samples of GaN grown by MOCVD and HVPE methods were prepared by Au evaporation. Ohmic contacts on both types of samples were prepared by rubbing the eutectic Al-Ga alloy. In the case of β -Ga₂O₃ it is difficult to find suitable samples because in the most of samples studied the recombination inside the depletion region cannot be neglected and under applied reverse bias the

induced current gain was observed. Therefore, the measurements were carried out on the Schottky barriers described in (Yakimov et al., 2021b).

Calculations of depth-dose dependencies for 4H-SiC, Au, Ni and semiconductors with deposited Au or Ni films were carried out using in-house Monte Carlo program based on the algorithm described in (Svintsov et al., 2022). As different definitions of electron range R exist in the literature (Reimer, 1998) in the present paper the definition proposed in (Werner et al., 1988) is used, i.e. R is equal to a thickness of layer within which 99 % of the total absorbed energy is transferred to the target.

3. Method for determining the average electron-hole pair ionization energy

The method based on fitting the collected current dependence on e-beam energy for the geometry with the electron beam perpendicular to the surface was proposed in (Wu and Wittry, 1978; Chi and Gatos, 1979) for diffusion length measurements in semiconductors. As shown in (Yakimov, 2015, 2021), this method allows to obtain the most correct values of excess carrier diffusion length in semiconductors with its submicron values. The collected current I_c under e-beam excitation for a Schottky barrier can be calculated as (Yakimov, 2015, 2021; Donolato, 1985)

$$I_c = \int_{t_m}^{\infty} \psi(z)g(z)dz \quad (1)$$

where t_m is the metal thickness, $g(z)$ is the depth dependent excess carrier generation rate (generation function) and $\psi(z)$ is the collection probability. As shown by Donolato (1985), the collection probability $\psi(z)$, which is the current induced by a unit charge at a depth z , can be obtained as a solution of the homogeneous diffusion equation with corresponding boundary conditions. For thick semiconductor structures the equation for $\psi(z)$ can be written as

$$D^2\psi(z) / dz^2 - \psi(z)/L^2 = 0 \quad (2)$$

with the boundary conditions $\psi(0) = 1$ and $\psi(z) \rightarrow 0$ at $z \rightarrow \infty$, where $L = (D\tau)^{0.5}$ is the excess carrier diffusion length, D is the ambipolar diffusivity equal to the minority carrier diffusivity at low excitation level and τ is the excess carrier lifetime. If L is independent of z and the recombination in the depletion region can be neglected, $\psi(z) = \exp[-(z-W)/L]$ for $z > W$ and $\psi(z) = 1$ for $z \leq W$, where W is the depletion region width. The generation function $g(z)$ is equal to the energy deposited at a depth z divided by E_i , i.e. $g(z) = U(E_b) \times h_s(z)/E_i$, where $U(E_b)$ is the total energy deposited inside a semiconductor and $h_s(z)$ is the normalized depth-dose

dependence in a semiconductor ($\int_0^{\infty} h_s(z)dz = 1$). Expression (1) can be

rewritten as

$$I_c(E_b) = \frac{U(E_b)}{E_i} \int_{t_m}^{\infty} \psi(z)h_s(z)dz \quad (3)$$

For the diffusion length measurements $\psi(z)$ is used as a fitting parameters, however, as seen from (3), $U(E_b)/E_i$ can also be obtained by the same fitting procedure. To obtain E_i only $U(E_b)$ should be calculated. Such way for E_i estimation does not need a well-calibrated detector to measure the deposited energy because beam current I_b is experimentally measured and the energy losses due to backscattering and absorption in the metal film can be calculated using the Monte Carlo method with a high precision. In addition, it allows to take into account the losses of excess carriers due to recombination in the quasi-neutral region by $\psi(z)$ fitting.

The $h_s(z)$ function for GaN and Ga₂O₃ has already been calculated by

the Monte Carlo algorithm in (Yakimov et al., 2007) and (Yakimov et al., 2018), respectively. To simplify the following fitting procedure, $h(z)$ as usually was approximated by the Gaussian functions that gave

$$h_s(z) = \frac{3.207}{R} \exp \left[-A \left(\frac{z}{R} - 0.11 \right)^2 \right] \quad (4)$$

where $R(\text{nm}) = 13.2 \times E_b(\text{keV})^{1.75}$ is the Bethe range and $A = \begin{cases} 42.8, z < 0.11 \cdot R \\ 16.5, z \geq 0.11 \cdot R \end{cases}$ for GaN (Yakimov et al., 2007). For Ga_2O_3

$$h_s(z) = \frac{1.603}{R} \exp \left[-A \left(\frac{z}{R} - 0.22 \right)^2 \right], \quad (5)$$

where $R(\text{nm}) = 7.34 \cdot E_b(\text{keV})^{1.75}$ и $A = \begin{cases} 12.86, z < 0.22 \cdot R \\ 3.97, z \geq 0.22 \cdot R \end{cases}$, for $\beta\text{-Ga}_2\text{O}_3$ (Yakimov et al., 2018).

For 4H-SiC the generation function was approximated in (Svintsov et al., 2018) with one Gaussian function. To improve the approximation precision in the present work it is approximated by two Gaussian functions as

$$h_s(z) = \frac{1.402}{R} \exp \left[-A \left(\frac{z}{R} - 0.33 \right)^2 \right], \quad (6)$$

where $R(\text{nm}) = 14.1 \cdot E_b(\text{keV})^{1.75}$ и $A = \begin{cases} 6.38, z < 0.33 \cdot R \\ 3.76, z \geq 0.33 \cdot R \end{cases}$.

Using these approximations for $h_s(z)$ the integral values in (3) can be obtained by fitting the experimental $I_c(E_b)$ dependences. The energy deposited inside a semiconductor can be calculated by subtracting the energy losses in the metal film from the energy absorbed in the entire structure

$$U(E_b) = E_b I_b (1 - \eta_E) \left[1 - \int_0^{t_m} h_m(z) dz \right], \quad (7)$$

where η_E is the backscatter energy coefficient of structure equal to the fraction of energy carried away by backscattered electrons and h_m is the normalized depth-dose dependence in a metal. As η in a common case depends on E_b (Niedrig, 1982; Hunger and Rogaschewski, 1986), this dependence should also be taken into account when fitting. For this purpose the $\eta_E(E_b)$ dependences for the structures studied should be calculated. It should be noted that in the vast majority of studies, the electron backscattering coefficient η was measured and calculated, although when using a semiconductor detector, the backscatter energy was measured rather than the number of backscattered electrons. For a thin film on bulk substrate η changes with E_b increase from its value for the film to that of substrate (Niedrig, 1982; Hunger and Rogaschewski, 1986). As shown in (Hunger and Rogaschewski, 1986) this dependence can be approximated as

$$\eta = \eta_s + b \times (\eta_F - \eta_s) \quad (8)$$

$$0 \leq b \leq 1$$

As our calculations have shown, a similar expression can be used to describe the $\eta_E(E_b)$ dependence

$$\eta_E = \eta_{Em} - (\eta_{Em} - \eta_{Esub}) \times \exp \left[-\frac{t_m}{a \times R_m \times E_b^{1.75}} \right], \quad (9)$$

where η_E , η_{Em} and η_{Esub} are the backscatter energy coefficients for the Schottky barrier, metal film and semiconductor substrate, respectively. The parameter a is equal to 0.22 and 0.226 for Au and Ni, respectively, and the electron range in a metal R_m can be expressed as $R_m(\text{nm}) = 2.3 \cdot E_b(\text{keV})^{1.75}$ for Au and $5.3 \cdot E_b(\text{keV})^{1.75}$ for Ni. For all materials used in the present study the beam energy dependence of backscatter energy coefficient can be approximated as

$$\eta_E = A + B/\exp(E_b/C) \quad (10)$$

where A , B and C coefficients for 4H-SiC, GaN, Ga_2O_3 , Ni and Au are shown in Table 1. As an example, the $\eta(E_b)$ dependence calculated by Monte Carlo method and that obtained using (9) for the Au Schottky barrier on 4H-SiC are presented in Fig. 1. It is seen that in spite of the large difference in the backscatter energy coefficients between 4H-SiC and Au Ex. (9) describes the dependence of η on E_b rather well.

As shown in (Svintsov et al., 2018), for a thin metal thickness $\int_0^{t_m} h_m(z) dz$ can be approximated by $\int_0^{t_{ef}} h_s(z) dz$, where $t_{ef} = t_m \times \rho_m/\rho_s$, ρ_m

and ρ_s are the densities of metal and semiconductor, respectively. Thus, all parameters on the right-hand side of expression (3) can be calculated, except for E_i and fitting the measured dependence $I_c(E_b)$ allows to obtain E_i , as a fitting parameter.

4. Results and discussion

4.1. 4H-SiC

Typical normalized $I_c(E_b)$ dependences measured on two Schottky barriers on 4H-SiC with different dopant concentration and diffusion length values are shown in Fig. 2. The corresponding simulated dependences are shown by magenta lines. For the Schottky barrier on a heavily doped crystal presented in Fig. 2 the fitting parameters are: Au thickness $t_m = 11$ nm, $W = 25$ nm, $L = 200$ nm, $E_i = 8.06$ eV; for a lightly doped crystal $t_m = 25$ nm, $W = 1.5$ μm , $L = 30$ μm , $E_i = 7.9$ eV. The value of E_i averaged over all measured 4H-SiC structures is of 8.04 ± 0.13 eV. For a comparison, the most recent reported values are 7.83 ± 0.02 eV (Gspomer et al., 2024) and 7.82 ± 0.02 eV (Garcia et al., 2013).

4.2. GaN

Typical normalized $I_c(E_b)$ dependences measured on initial and irradiated with 6 MeV electrons with a dose of $1.1 \times 10^{16} \text{cm}^{-2}$ Schottky barriers are shown in Fig. 3. For the Schottky barrier on the initial sample the fitting parameters are: $t_m = 16$ nm, $W = 295$ nm, $L = 320$ nm, $E_i = 10.3$ eV. For the Schottky barrier on the irradiated sample $t_m = 12$ nm, $W = 230$ nm, $L = 80$ nm, $E_i = 10.8$ eV. The value of E_i averaged over all GaN structures is of 10 ± 0.5 eV. The values of $E_i = 9.51$ eV and $E_i = 9.59$ eV, which we obtained earlier (Yakimov, 2023; Yakimov et al., 2021a; Svintsov et al., 2018) fall within this range. However, lower value of 8.9 eV was reported in (Sellin and Vaitkus, 2006). The spread in E_i values in GaN is significantly larger than that in 4H-SiC, and the reasons for this are not yet clear. However, if it is determined by the excess carrier recombination inside the depletion region, which was neglected in the model, then the smaller E_i values are more correct. It should be also noted that from the Klein prediction (Klein, 1968) E_i should be in the range from 10 to 10.5 eV.

4.3. $\beta\text{-Ga}_2\text{O}_3$

Measurements of E_i in $\beta\text{-Ga}_2\text{O}_3$ are associated with certain difficulties. Previous studies of Schottky barriers formed on a large number of $\beta\text{-Ga}_2\text{O}_3$ crystals and epitaxial films using the EBIC method have

Table 1
Empirical coefficients for the materials studied.

	A	B	C
4H-SiC	0.0836	0.037	7.95
GaN	0.216	0.071	6.58
$\beta\text{-Ga}_2\text{O}_3$	0.204	0.068	5.99
Ni	0.22	0.08	8
Au	0.447	0.17	4.83

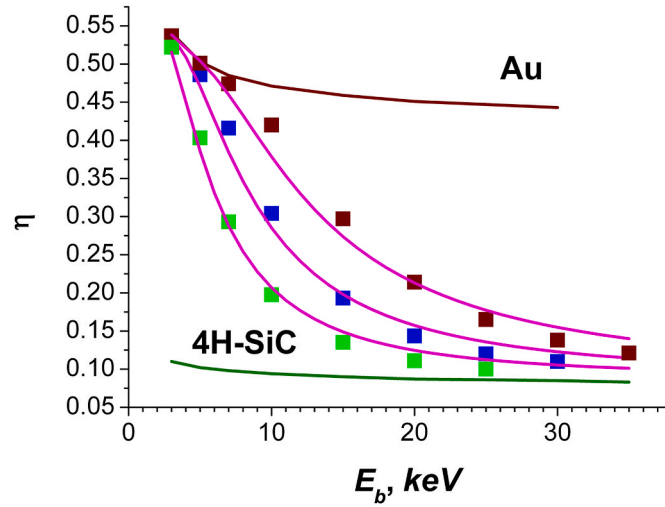


Fig. 1. Calculated $\eta(E_b)$ dependences using the Monte Carlo algorithm and those obtained using Expression (9) (magenta lines) for the Au Schottky barrier on 4H-SiC with the Au film thickness of 10 (green symbols), 20 (blue symbols) and 40 nm (wine symbols). The dependences for Au and 4H-SiC are also shown with magenta lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

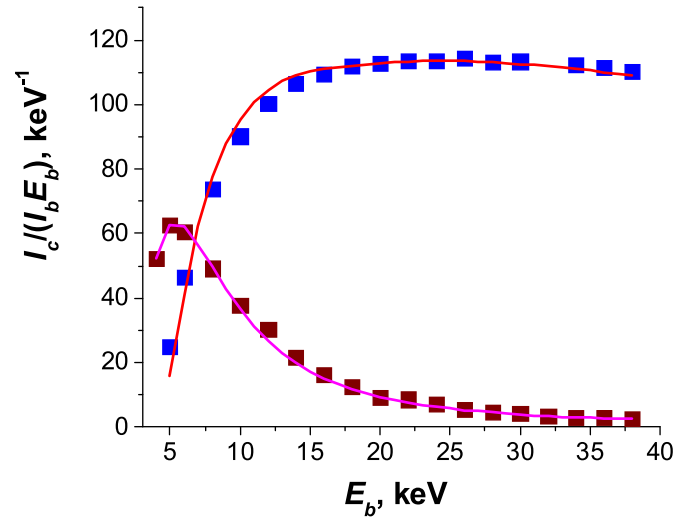


Fig. 2. Normalized $I_c(E_b)$ dependences for heavily (wine symbols) and lightly doped 4H-SiC (blue symbols). Simulated dependences are shown by magenta lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

shown (Yakimov et al., 2020, 2021a) that without an applied bias the collected current is significantly lower than the expected value. Moreover, it increases with applied reverse bias even at beam energies, at which the excess carriers are created inside the depletion region. This decrease in the collection efficiency can be determined by hole trapping on the surface states near the interface with the Schottky diode metal and/or by partial hole self-trapping in the quasi-neutral region (Varley et al., 2012; Deák et al., 2017; Frodason et al., 2020). An increase of the reverse bias seems to suppress the both effects, however, in the most of studied Schottky barriers the induced current gain was observed at reverse bias (Yakimov et al., 2021b; Polyakov et al., 2025). Fortunately, on one of the β -Ga₂O₃ epitaxial films it was possible to form barriers, in which the collected current induced by a beam with energy of 4 keV increased at low bias, and then remained practically unchanged up to voltages of about 80 V. Under an assumption that in the bias independent region the hole recombination inside the depletion region is suppressed the E_i value can be estimated from the $I_c(E_b)$ dependence. As an example, such dependence measured at 5 V reverse bias is shown in Fig. 4 together with the simulated curve. The fitting parameters used are: $t_m = 22$ nm, $L = 230$ nm, $E_i = 14.8$ eV. The E_i obtained is smaller

than that obtained in (Yakimov et al., 2021a) on the similar structure but larger than 11.24 eV obtained by solving the semiclassical Boltzmann transport equation (Nielsen and Fischetti, 2023). However, it should be taken into account that in (Yakimov et al., 2021a) the estimation was made using the collected current measured at 4 keV. In the present work the full $I_c(E_b)$ dependence is fitted. Additionally, the dependence of the energy carried away by the backscattered electrons on the beam energy is taken into account. Therefore, it can be expected that the obtained E_i value is more accurate than that obtained in (Yakimov et al., 2021a). It should be also noted that according the Klein prediction (Klein, 1968) E_i should be in the range from 13.95 to 14.45 eV, which correlates with the value obtained in the present work.

5. Conclusion

Thus, the method for the measurements of average energy consumed in the creation of an electron-hole pair in semiconductors based on fitting the collected current dependence on beam energy has been described. The effect of metal film on the backscatter energy is estimated for 4H-SiC, GaN and β -Ga₂O₃ Schottky barriers. The empirical

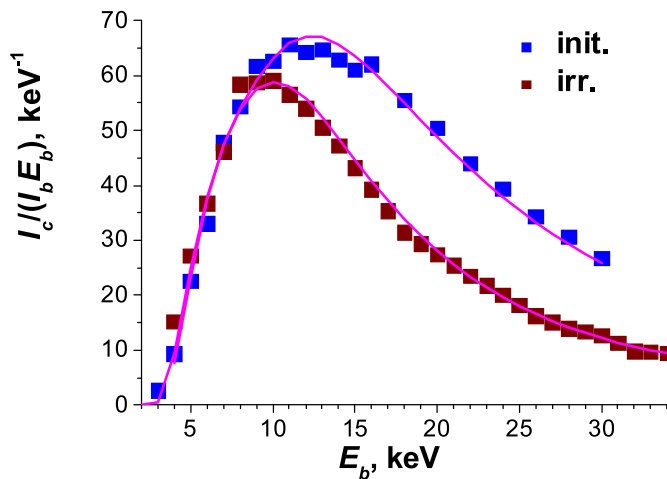


Fig. 3. Normalized $I_c(E_b)$ dependences for the initial and electron irradiated Au/GaN Schottky barriers. Simulated dependences are shown by magenta lines.

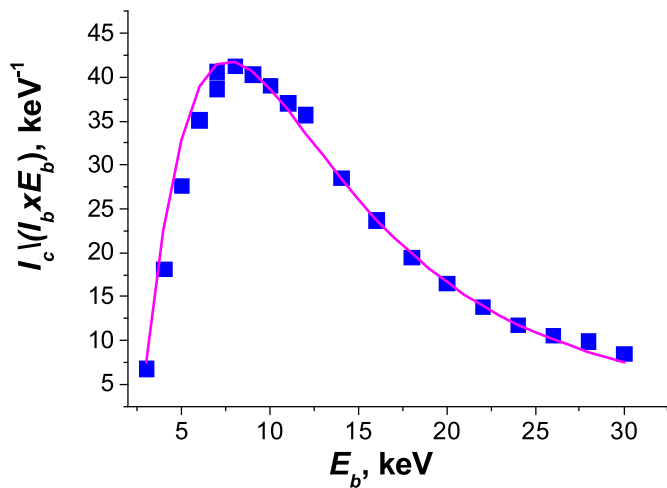


Fig. 4. Normalized $I_c(E_b)$ dependences for the Ni/ β -Ga₂O₃ Schottky barrier measured at reverse bias of 5 V. Simulated dependence is shown by the magenta line.

expression describing the dependence of this energy on beam energy is proposed. The approximated depth-dose dependence for 4H-SiC is obtained by fitting the dependence calculated using the Monte Carlo algorithm. Values of average electron-hole pair ionization energy are estimated for 4H-SiC, GaN and β -Ga₂O₃ taking into account the beam energy absorption in the metal film and its effect on the electron backscattering. The values obtained by the method discussed are 8.04 ± 0.13 , 10 ± 0.5 eV and 14.8 eV for 4H-SiC, GaN and β -Ga₂O₃, respectively.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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